Regression Rate Analysis of Hybrid Rocket Motors Using HTPB-Based Fuels: A Comparative Study of Gaseous Oxygen and Nitrous Oxide as Oxidizers

COURSE CODE AND COURSE

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May, 2025

CERTIFICATE

Certified that this Seminar Report titled Regression Rate Analysis of Hybrid Rocket Motors Using HTPB-Based Fuels: A Comparative Study of Gaseous Oxygen and Nitrous Oxide as Oxidizers by Darsh Butala is approved by me for submission. Certified further that, to the best of my knowledge, the report represents work carried out by the student.

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ABSTRACT

Hybrid rocket propulsion offers a unique balance between the controllability of liquid systems and the structural simplicity of solid motors. Among oxidizer options, nitrous oxide (N_2O) is increasingly favored for its self-pressurizing behavior, thermal stability, and ease of storage—attributes that simplify propulsion system design and improve safety. However, N_2O exhibits a lower fuel regression rate compared to gaseous oxygen (GOX), primarily due to its inert nitrogen content and lower flame temperatures.

This study reviews recent advancements in enhancing the regression rate of hydroxyl-terminated polybutadiene (HTPB) fuels through the inclusion of nano-sized energetic additives, particularly aluminum-based formulations. Experimental findings indicate that nano-aluminum additives significantly improve heat feedback mechanisms, thereby increasing regression rates by up to 25% under moderate mass flux conditions. Additionally, fluoropolymer-coated nano-aluminum further augments ignition efficiency and energy release while maintaining favorable mechanical properties.

The combined approach of energetic additives and grain geometry optimization bridges the performance gap between N_2O and GOX systems. This paper concludes that N_2O-based hybrid rockets, when enhanced with aluminized HTPB, represent a practical and efficient solution for tactical, educational, and suborbital missions. Future improvements in additive formulations and decomposition strategies are poised to further enhance the viability of N_2O-powered hybrid propulsion.

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NOMENCLATURE

Symbol	Description	Unit
$\dot{r}$	Fuel regression rate	mm/s
G_{ox}	Oxidizer mass flux	kg/m^2·s
ρ_f	Solid fuel density	kg/m^3
ΔH	Effective heat of gasification	J/kg
$\dot{Q}_w$	Wall heat flux	W/m^2
C_H	Stanton number with blowing	–
C_{H0}	Baseline Stanton number	–
Re_x	Reynolds number ($\rho_e u_e x/\mu$)	–
ρ_e	Density at boundary layer edge	kg/m^3
u_e	Velocity at boundary layer edge	m/s
h_e	Enthalpy at flame position	J/kg
h_w	Enthalpy at wall	J/kg
B	Blowing parameter	–
η_c	Velocity profile exponent	–
μ	Dynamic viscosity	Pa · s
a, n	Empirical regression rate coefficients	–
C	Turbulent flow constant	–
u_c	Characteristic combustion velocity	m/s

Subscripts/Superscripts

e	Boundary layer edge condition
w	Wall condition
c	Combustion characteristic value
ox	Oxidizer-related quantity
f	Fuel-related quantity
$\sim$	Proportional relationship
$\cdot$	Time derivative

Chapter 1

INTRODUCTION

Hybrid rocket propulsion systems, which utilize a solid fuel in conjunction with a liquid or gaseous oxidizer, offer a compelling middle ground between conventional solid and liquid propulsion technologies. Initially conceptualized in the early 1930s, hybrid propulsion leverages the storage simplicity and structural integrity of solid fuels—such as hydroxyl-terminated polybutadiene (HTPB)—with the controllability and throttleability of liquid or gaseous oxidizers, including liquid oxygen (LOX), gaseous oxygen, or nitrous oxide (N_2O)[1]. While these systems present notable advantages for space launch vehicles, upper-stage propulsion, and tactical missile applications, their widespread adoption has been hindered by persistent technical challenges. Nevertheless, recent research and technological advancements have renewed interest in hybrid propulsion architectures.

ADVANTAGES OF HYBRID PROPULSION SYSTEMS

IMPROVED SAFETY CHARACTERISTICS

One of the principal advantages of hybrid propulsion lies in the inherent safety of the system. The physical separation of fuel and oxidizer significantly reduces the risk of unintentional detonation, distinguishing hybrids from fully solid or liquid systems. Furthermore, solid fuel grains are less susceptible to structural flaws; cracks or voids do not typically result in catastrophic combustion events. In their unloaded state, hybrid propellants are generally inert, which simplifies handling, transportation, and storage when compared to hypergolic liquids or premixed solid propellants[1].

OPERATIONAL FLEXIBILITY

Unlike solid motors, hybrid rockets allow for real-time control over thrust output through modulation of oxidizer flow rates. This throttleability, along with the ability to execute multiple ignition sequences, facilitates mission-critical capabilities such as abort maneuvers and trajectory corrections. These attributes make hybrids particularly suitable for reusable launch vehicles (RLVs) and crewed missions requiring precision and safety[1].

COST AND DESIGN EFFICIENCY

Hybrid systems benefit from reduced mechanical complexity compared to liquid bipropellant engines. The elimination of turbopumps and simplification of plumbing infrastructure lead to decreased manufacturing and operational costs. While hardware costs are comparable to solid rocket systems, the propellant costs align more closely with those of liquid propulsion systems, making hybrids an economically viable option for certain mission profiles[1].

DISADVANTAGES AND TECHNICAL CHALLENGES OF HYBRID PROPULSION SYSTEM

LOW REGRESSION RATES OF SOLID FUELS

A central limitation of hybrid propulsion is the low regression rate of conventional solid fuels like HTPB, typically on the order of 1–3 mm/s. This necessitates the use of complex multiport grain configurations to generate adequate thrust. Emerging fuel formulations based on paraffin wax exhibit up to four times higher regression rates; however, they often require mechanical reinforcements to prevent grain deformation. Enhancements such as the inclusion of ultrafine aluminum particles have demonstrated regression rate increases of approximately 40% due to intensified near-surface exothermic reactions[1].

SUBOPTIMAL COMBUSTION EFFICIENCY

Hybrid motors generally exhibit lower combustion efficiency—typically between 93% and 98%—compared to traditional solid or liquid systems. The primary cause is diffusion-limited mixing at the fuel-oxidizer interface, which impairs complete energy release. Additionally, incomplete fuel consumption often results in unburned "slivers" remaining within the combustion chamber, thereby reducing mass utilization. The thermal feedback mechanisms, governed by both radiative and convective heat transfer, further complicate the combustion process[1].

OXIDIZER-TO-FUEL RATIO INSTABILITIES

As combustion progresses, the internal port geometry dynamically evolves, leading to fluctuations in the oxidizer-to-fuel (O/F) ratio. These variations directly impact the specific impulse (I_{sp}), leading to performance instability over the burn duration. To counter this, advanced grain shaping and oxidizer injection techniques are required to maintain a more consistent burn profile[1].

CHALLENGES IN SYSTEM SCALING

Unlike solid rockets, the performance of hybrid propulsion systems does not scale linearly due to the dependence of regression rate on local mass flux, port geometry, and complex heat transfer dynamics. This makes reliable scaling from subscale testbeds to full-scale flight systems particularly challenging. Accurate predictive modeling of internal ballistics is essential and typically necessitates extensive empirical validation and computational simulations[1].

Chapter 2

COMMON TYPES OF FUEL AND OXIDIZER USED IN HYBRID ROCKETS

The following is a selection of commonly utilized fuel and oxidizer combinations for hybrid rocket motors, based on the data presented by Estey and Whittinghill (1992) in their comprehensive evaluation of hybrid propellant alternatives.

Table 2.1: Comparison of Hybrid Rocket Fuel Properties

Fuel Type	**Density (g/cm^3)**	**Heat of Formation (cal/g)**	**Advantages**	**Limitations**
HTPB	0.94	-18.5	Cost-effective; Non-toxic; Mechanically robust; Allows complex grain geometries	Moderate regression rate; Requires oxidizer mass flux optimization
HTPB + 20% Aluminum	1.11	-17.8	Higher energy density; Improves regression rate (20%); Better volumetric efficiency	Aluminum oxide residue may erode nozzle; Advanced blends are costly and complex
PMMA	1.58	-906	Transparent for diagnostics; Non-toxic combustion products	Lower specific impulse

Oxidizer	Boiling Point (°C)	Density (g/cm³)	Advantages	Limitations
Oxygen (O2)	–183	1.14	High performance; Environmentally friendly	Requires cryogenic storage and boil-off control if planned to use as LOX
Nitrous Oxide (NO)	–89	0.79	Self-pressurizing; Easy handling	Lower density and performance compared to LOX
Nitrogen Tetroxide (NO)	+21	1.45	Storable at room temperature	Highly toxic and corrosive
FLOX (30% F + 70% LOX)	–188	1.24	Highest Isp; Suitable for high-V missions	Highly reactive and expensive

Table 2.2: Comparison of commonly used oxidizers in hybrid rocket propulsion

PRACTICAL APPLICATIONS AND DESIGN IMPLICATIONS

- **Educational and Experimental Systems:** HTPB/N_2O systems are ideal for university and amateur rocketry due to their self-pressurizing behavior and ease of handling.
- **Operational Launch Stages:** LOX/HTPB systems, especially when metalized, provide a balance between specific impulse ($\approx$ 360 s), grain integrity, and operational safety.
- **High-Energy Missions:** FLOX and lithium-enhanced HTPB fuels enable I_{sp} values exceeding 400 s, making them attractive for interplanetary stages, albeit with higher safety and financial constraints.

Chapter 3

COMPARATIVE ANALYSIS OF NITROUS OXIDE AND GASEOUS OXYGEN

Having reviewed several commonly employed fuel and oxidizer combinations for hybrid rocket propulsion, this study now turns its focus to a comparative analysis between nitrous oxide (N_2O) and gaseous oxygen (O_2) as oxidizers. The objective is to evaluate their respective thermophysical and operational characteristics and to illustrate circumstances under which N_2O may offer distinct advantages over gaseous O_2.

Below Table presents fundamental properties of N_2O and O_2, including density, boiling point, and other key parameters, with data compiled from the NIST Chemistry WebBook and Air Liquide Gas Encyclopedias[3,4,5,6].

Table 3.1: Comparison of Key Properties of Oxygen (O_2) and Nitrous Oxide (N_2O)

Property	**Oxygen (O_2)**	**Nitrous Oxide (N_2O)**
Chemical Formula	O_2	N_2O
Molar Mass (g/mol)	31.9988	44.0128
Triple Point (K, bar)	54.33 K, 1.5×10^{-3} bar	182.43 K, 0.8789 bar
Melting Point (°C)	–218.35	–90.86
Boiling Point (°C)	–182.96	–88.48
Latent Heat of Vaporization (kJ/kg)	213.05	374.3
Critical Temperature (K)	154.58	309.56
Critical Pressure (bar)	50.43	72.38
Critical Density (kg/m^3)	435.95	451.9
Liquid Density at BP (kg/m^3)	1141.2	1230.5
Gas/Liquid Volume Ratio at BP	799	676
$\Delta_f H^\circ_{298}$ (kJ/mol)	0 (reference)	+82.05
Standard Molar Entropy S°_{298} (J/mol·K)	205.14	219.96
Solubility in Water at 25°C (mol/mol)	2.29×10^{-5}	4.386×10^{-4}
Vapor Pressure at 25°C (bar)	–	57.29
Material Compatibility (Metals)	Stainless steel, aluminum	Stainless steel, aluminum
Material Compatibility (Polymers)	PTFE; avoid PVC, nitrile	PTFE; avoid PVC, hydrocarbons
Safety Notes	Supports combustion; reactive	Decomposes exothermically; oxidizer

ANALYSIS OF NITROUS OXIDE AND GASEOUS OXYGEN IN HTPB-BASED HYBRID ROCKETS

This section provides a technical comparison between N_2O and O_2 as oxidizers in HTPB-based hybrid rockets, focusing on aspects such as system complexity, storage requirements, safety, and performance implications.

ADVANTAGES OF NITROUS OXIDE OVER OXYGEN

Self-Pressurization Capability

Nitrous oxide exhibits a saturated vapor pressure of approximately 50 bar at room temperature (~20°C), which allows it to self-pressurize the propulsion feed system. This eliminates the need for auxiliary pressurization subsystems such as helium bottles or turbopumps, significantly simplifying the propulsion architecture and reducing failure points compared to cryogenic LOX systems that require external pressurization support[7].

Ambient-Temperature Storage

Unlike LOX, which requires cryogenic storage infrastructure and boil-off mitigation strategies, N_2O remains in a liquid state at ambient temperatures. This property minimizes ground-support equipment complexity and enables easier transportation and long-term storage—factors especially advantageous for small-scale, experimental, or tactical missions[7].

Improved Handling Safety

Nitrous oxide is relatively less hazardous to handle under ambient conditions compared to gaseous or liquid oxygen. While still an oxidizer, N_2O poses lower risks of spontaneous ignition or material embrittlement and is non-cryogenic and non-toxic, making it favorable for academic, amateur, and suborbital launch platforms[7].

Volumetric Efficiency

Though N_2O-based systems offer lower specific impulse than LOX systems, their higher liquid density (approximately 1.22 g/cm^3 at 20°C) and self-pressurization result in high impulse-per-tank-volume ratios. This makes N_2O attractive for missions constrained by volume rather than total impulse[7].

LIMITATIONS OF NITROUS OXIDE COMPARED TO OXYGEN

Lower Specific Impulse (I_{sp}Isp)

N_2O/HTPB hybrid motors typically achieve vacuum specific impulse values in the range of 230–250 seconds, which is significantly lower than LOX/HTPB systems that can exceed 275–300 seconds. This discrepancy arises from N_2O's lower adiabatic flame temperature and its endothermic decomposition, which reduces thermal energy release and combustion efficiency[8].

Reduced Regression Rates

The endothermic decomposition of N_2O delays energy transfer to the HTPB fuel grain, resulting in slower regression rates—typically 20–30% lower than those observed in LOX-fed systems[8].

Thermal and Chemical Stability Risks

Although generally stable, N_2O can undergo exothermic decomposition (N2O -> N2 + 1/2 O2, $\Delta H = -82$ kJ/mol) when exposed to elevated temperatures, reactive surfaces, or contaminants. This behavior introduces potential instability risks during prolonged storage or under abnormal thermal conditions, necessitating careful selection of construction materials and rigorous system cleanliness[7].

Nitrous oxide offers a compelling oxidizer option for hybrid rocket systems targeting applications where simplicity, safety, and operational flexibility are prioritized over peak performance. Its self-pressurizing nature, ambient storability, and ease of handling make it ideal for sounding rockets, tactical suborbital vehicles, and academic missions. Conversely, liquid or gaseous oxygen remains superior for high-performance and orbital-class missions due to its higher energy content and specific impulse.

Building upon the understanding of various fuels and oxidizers, as well as the thermochemical properties of nitrous oxide (N_2O) and gaseous oxygen (O_2), this paper now delves into the detailed mechanism of regression rate in hybrid rockets and the derivation of the Marxman equation.

Chapter 4

COMBUSTION MECHANISM IN HYBRID ROCKET ENGINES

The combustion process in hybrid rocket engines is governed by the interaction between a gaseous oxidizer and a solid fuel surface, resulting in a heterogeneous combustion regime. The model describing this behavior is the Marxman equation, which captures the fuel regression rate under turbulent boundary layer flow conditions.

The model was first introduced by Marxman, Wooldridge, and Muzzy in the *AIAA Heterogeneous Combustion Conference* (1963). Their research provided the foundational understanding of turbulent boundary layer combustion in hybrid propulsion systems.

MARXMAN EQUATION FOR FUEL REGRESSION RATE

The Marxman equation relates the fuel regression rate $\dot{r}$ to the oxidizer mass flux G_{ox} as:

$$\dot{r} = a \cdot G_{\text{ox}}^{n} \tag{4.1}$$

where:

- $\dot{r}$: Regression rate of the solid fuel (mm/s)
- G_{ox}: Oxidizer mass flux at the fuel surface (kg/m^2·s)
- a, n: Empirical coefficients dependent on flow conditions, fuel properties, and chamber pressure

The derivation of the equation presented here is based on the works of Marxman, G. A., Wooldridge, C. E., & Muzzy, R. J. (1963), *Fundamentals of Hybrid Boundary Layer Combustion*, and Marxman, G. A., & Gilbert, M. (1963), *Turbulent Boundary Layer Combustion in the Hybrid Rocket*, United Technology Center, AIAA.

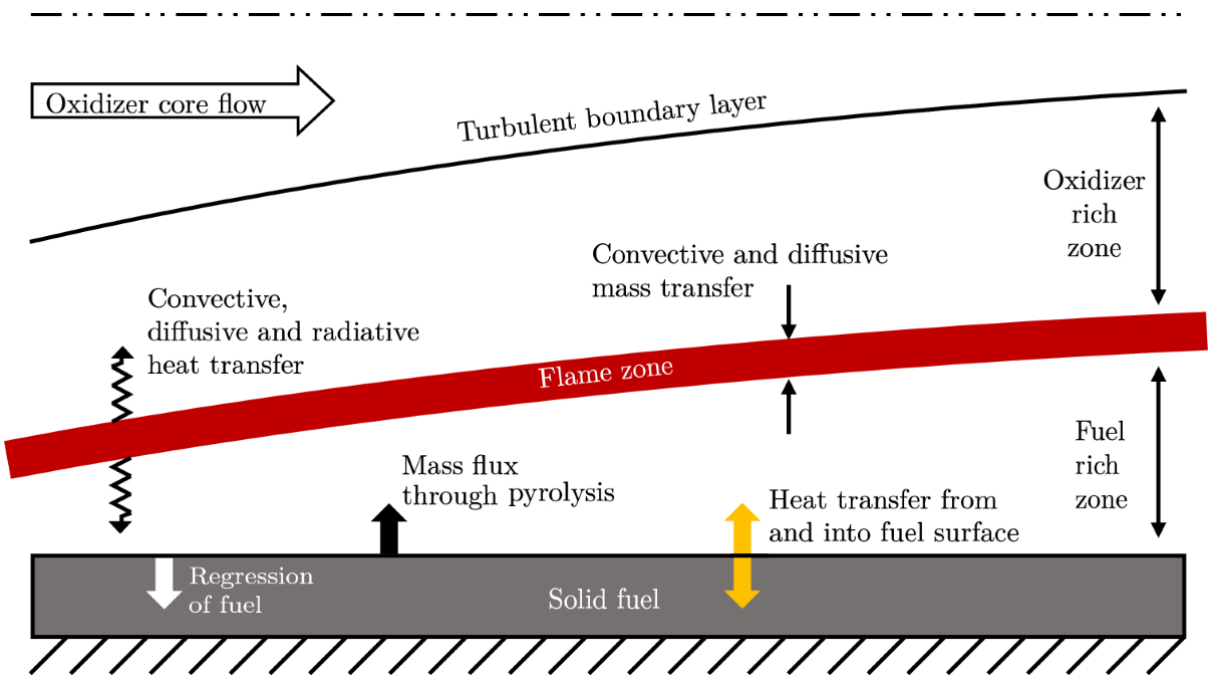

Figure 4.1: Regression Mechanism in Hybrids[18]

STEP-BY-STEP DERIVATION OF THE HYBRID ROCKET REGRESSION RATE EQUATION

This derivation follows the classical Marxman theory to determine the regression rate ($\dot{r}$) in a hybrid rocket motor, focusing on the energy balance and turbulent boundary layer heat transfer.

STEP 1: ENERGY BALANCE AT THE FUEL SURFACE

The regression rate $\dot{r}$ is governed by the energy required to gasify the solid fuel:

$$\rho_f \dot{r} \Delta H = \dot{Q}_w \tag{4.2}$$

where:

- ρ_f: fuel density (kg/m^3)
- $\dot{r}$: regression rate (m/s)
- ΔH: effective heat of gasification (J/kg)
- $\dot{Q}_w$: heat flux to the wall (W/m^2)

Solving for $\dot{r}$:

$$\dot{r} = \frac{\dot{Q}_w}{\rho_f \Delta H} \tag{4.3}$$

STEP 2: CONVECTIVE HEAT FLUX IN A TURBULENT BOUNDARY LAYER

The heat flux can be expressed using the Stanton number C_H:

$$\dot{Q}_w = C_H \cdot \rho_e u_e (h_e - h_w) \tag{4.4}$$

where:

- $\rho_e u_e = G$: oxidizer mass flux (kg/m^2·s)
- h_e: enthalpy at the flame
- h_w: enthalpy at the wall

Substituting into the regression rate expression:

$$\dot{r} = \frac{C_H \rho_e u_e (h_e - h_w)}{\rho_f \Delta H} \tag{4.5}$$

STEP 3: STANTON NUMBER FOR TURBULENT FLOW WITH BLOWING

Blowing from fuel gasification reduces the Stanton number:

$$C_H = \left(\frac{C_H}{C_{H0}}\right) C_{H0} \tag{4.6}$$

where:

- C_{H0}: Stanton number without blowing
- $\frac{C_H}{C_{H0}}$: blowing correction factor (typically ∼0.1 for hybrid rockets)

The baseline Stanton number in turbulent flow is given by:

$$C_{H0} = C \cdot \mathrm{Re}_x^{-0.2} \tag{4.7}$$

with:

- $C \approx 0.036$: empirical constant
- $\mathrm{Re}_x = \dfrac{\rho_e u_e x}{\mu}$: local Reynolds number

Thus, the corrected Stanton number becomes:

$$C_H = \left(\frac{C_H}{C_{H0}}\right) C \cdot \mathrm{Re}_x^{-0.2} \tag{4.8}$$

STEP 4: VELOCITY RATIO u_e/u_c

Due to flame standoff, the actual velocity at the flame u_c is less than u_e:

$$\frac{u_c}{u_e} \approx \frac{\eta_c^n \left(1 + \frac{1}{2} B \eta_c^n\right)}{1 + \frac{1}{2} B} \tag{4.9}$$

Using typical values ($B \approx 10$, $\eta_c = 0.15$, $n = 1/7$):

$$\frac{u_e}{u_c} \approx 1.7 \tag{4.10}$$

STEP 5: COMBINE ALL TERMS

Substituting C_H, u_e/u_c into the regression rate:

$$\dot{r} = \frac{\left(\frac{C_H}{C_{H0}}\right) C \cdot \mathrm{Re}_x^{-0.2} \cdot \rho_e u_e (h_e - h_w)}{\rho_f \Delta H} \cdot \left(\frac{u_e}{u_c}\right) \tag{4.11}$$

Simplifying:

$$\dot{r} = \frac{C \cdot G \cdot \mathrm{Re}_x^{-0.2}}{\rho_f} \cdot \left(\frac{C_H}{C_{H0}}\right) \cdot \left(\frac{u_e}{u_c}\right) \cdot \left(\frac{h_e - h_w}{\Delta H}\right) \tag{4.12}$$

Chapter 5

SCOPE OF EXISTING RESEARCH AND MOTIVATION FOR CURRENT STUDY

While extensive research has been conducted on the regression behavior of hydroxyl-terminated polybutadiene (HTPB) in combination with gaseous oxygen (GOX)—particularly with various performance-enhancing additives—comparative studies involving nitrous oxide (N_2O) as the oxidizer remain relatively limited. The current work aims to consolidate and critically review the body of literature related to HTPB/GOX hybrid systems with additives, and based on established combustion mechanisms and trends, the outcomes if N_2O were used as a substitute oxidizer are presented.

Furthermore, all available and reliable experimental studies involving HTPB/N_2O combinations with different additives have also been included to strengthen the comparative analysis. This dual approach not only highlights the knowledge gap in N_2O-based hybrid research but also provides a foundation for future experimental and numerical investigations.

PERFORMANCE ENHANCEMENT OF HTPB-BASED HYBRID ROCKET FUELS: INFLUENCE OF GEOMETRY AND NANO-SIZED ENERGETIC ADDITIVES

Hybrid rocket motors employing hydroxyl-terminated polybutadiene (HTPB) as fuel exhibit combustion characteristics that are strongly influenced by both geometric parameters and compositional modifications. Recent experimental investigations provide key insights into how port configuration, grain length, and nano-sized energetic additives synergistically affect regression behavior and overall propulsion performance.

AXIAL REGRESSION BEHAVIOR AND FLAME STRUCTURE

The regression rate in HTPB/GOX hybrid systems is observed to increase along the axial direction of the grain, particularly due to the rise in local oxidizer mass flux that enhances convective heat transfer. However, in longer grains (e.g., 545 mm), this trend attenuates toward the aft end as the system becomes fuel-rich, leading to reduced flame

temperatures and diminished heat feedback[11,12].

This axial non-uniformity in regression is directly tied to the oxidizer-to-fuel (O/F) ratio, which varies from ~ 4.5 in short grains (fuel-lean) to ~ 2.5 in longer grains (fuel-rich), thereby influencing flame structure and combustion efficiency.

PRESSURE-INDEPENDENT REGRESSION AND MODIFIED SCALING LAWS

Regression rates of HTPB have been found largely insensitive to chamber pressure in the 2.0–3.0 MPa range, consistent with diffusion-limited boundary-layer combustion theory. Contrary to the classical Marxman model that predicts a dependence $\tilde{r} \propto G^{0.8}$, empirical correlations suggest reduced mass flux sensitivity and a pronounced dependence on port diameter. For example:

- **Pure HTPB:**

$$\tilde{r} = 6.37 \times 10^{-5}\, G_o^{0.41} D^{-0.34} \tag{5.1}$$

- **HTPB + AP:**

$$\tilde{r} = 1.88 \times 10^{-5}\, G_o^{0.42} D^{-0.69} \tag{5.2}$$

These deviations point to additional mechanisms—such as increased flame radiation and turbulent mixing—contributing to fuel regression[12].

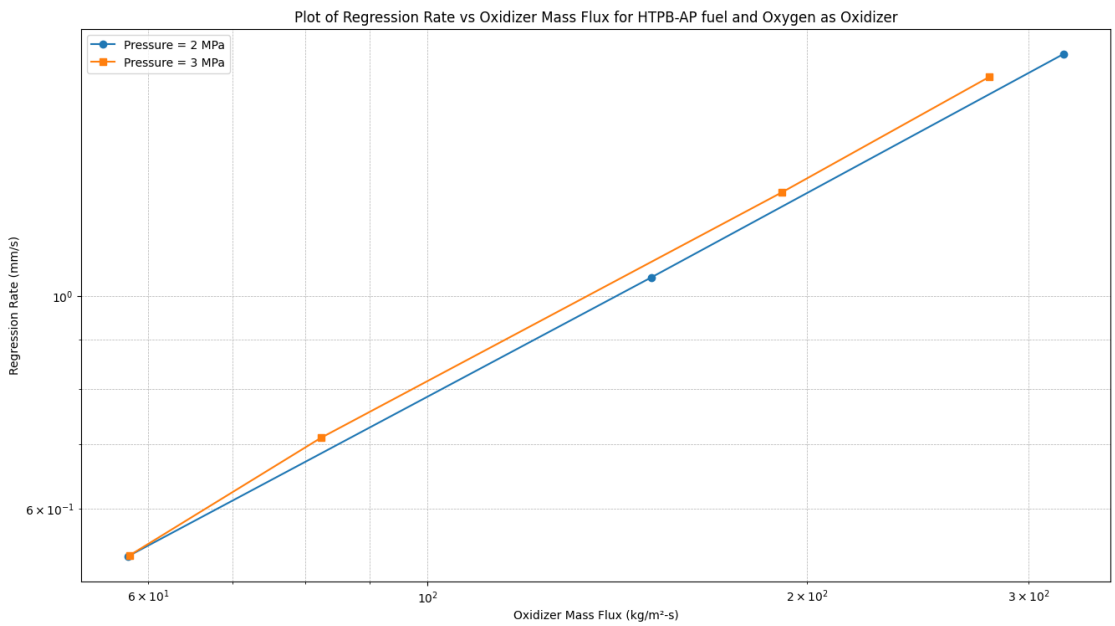

Figure 5.1: Plot of Regression Rate vs Oxidizer Mass Flux for HTPB-AP fuel and Oxygen as Oxidizer[12]

REGRESSION RATE AUGMENTATION STRATEGIES

Port Diameter Reduction

Reduction of port diameter from 20 mm to 12 mm has been shown to nearly double the regression rate, attributed to increased oxidizer mass flux and steeper heat flux gradients. This geometric tuning outperformed even additive-based enhancements in certain configurations.

Energetic Additives

Incorporating energetic materials like ammonium perchlorate (AP) and aluminum (Al) into HTPB improves regression rates significantly. AP (7.55%) contributes to a 40–60% increase at high oxidizer mass fluxes (∼300 kg/m^2·s) due to gas-phase exothermic reactions, while micron-sized Al (19.67%) further elevates regression but necessitates erosion-resistant nozzle designs[12].

More recently, the integration of nano-sized aluminum particles has shown transformational potential. Risha et al.[15] reported a 120% increase in burning rate using Viton-A-coated nano-aluminum (C-Alex®). This is due to the particles' small diameter (e.g., ∼70 nm) enabling rapid energy release via the D^2-law, enhanced by uniform dispersion within the fuel matrix[14,15]. Fluoropolymer coatings facilitate ignition by releasing fluorine species that disrupt protective oxide layers on aluminum, achieving C^* efficiencies of up to 92% and outperforming uncoated flakes by 43%.

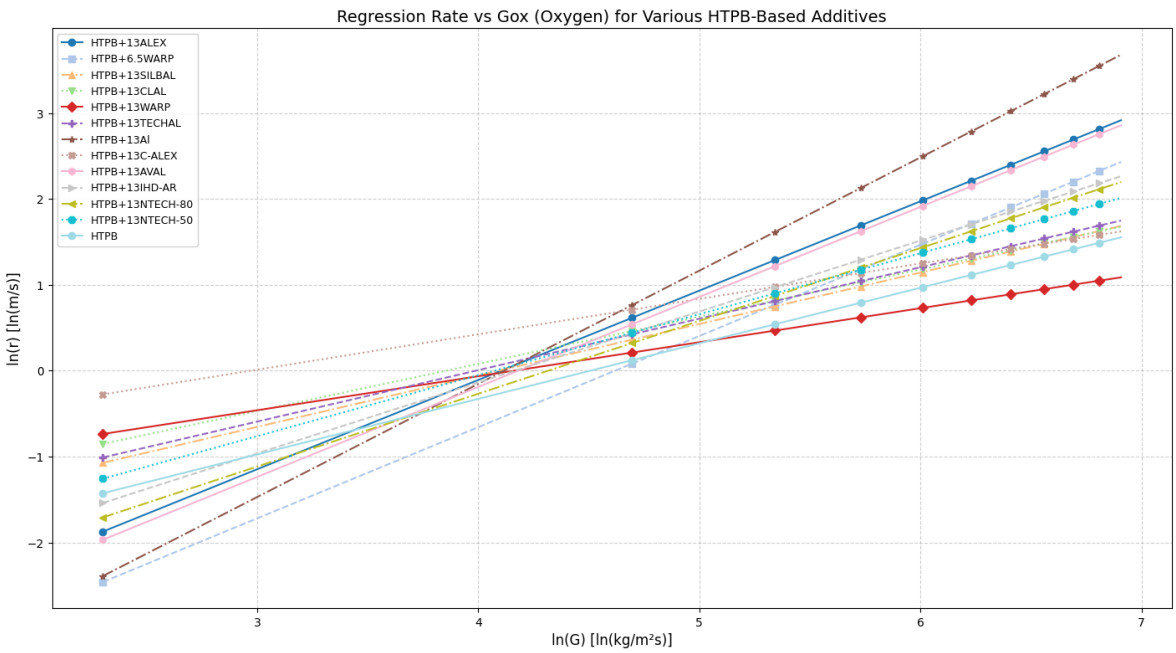

Figure 5.2: Regression rate of HTPB-based solid fuels with various energetic additives using gaseous oxygen (GOX) as the oxidizer.[13,14,15]

DESIGN IMPLICATIONS

Combining port diameter optimization with energetic additives yields synergistic benefits. Multi-perforated grains with reduced port diameters provide high surface area for combustion, enabling compact motor configurations with superior thrust-to-volume ratios. The dual strategy of geometric and compositional enhancement allows tailored performance for both tactical and high-altitude mission profiles.

OXIDIZER PERFORMANCE AND THE ROLE OF ALUMINUM ADDITIVES IN HTPB-BASED HYBRID FUELS

EFFECT OF OXIDIZER TYPE: GOX VS. N_2O

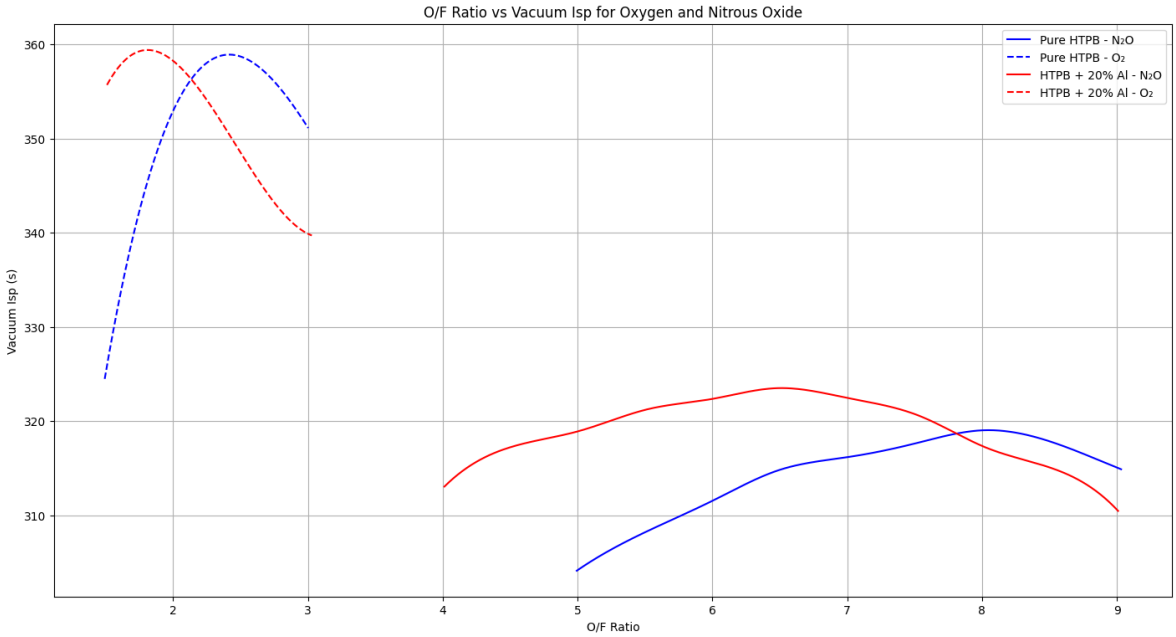

Figure 5.3: Vaccum Specific Impulse vs O/F ratio for Oxygen and N_2O using HTPB and with Aluminum additive[2]

The performance of HTPB-based hybrid rocket systems is significantly influenced by the choice of oxidizer. Comparative studies reveal that nitrous oxide (N_2O) reduces the average regression rate of pure HTPB by approximately 25% relative to gaseous oxygen (GOX), primarily due to the presence of inert nitrogen in N_2O that dilutes combustion gases and lowers flame temperatures[17].

Despite this decrease in regression performance, N_2O offers key operational advantages. These include self-pressurization at ambient temperatures, the elimination of cryogenic handling infrastructure, and reduced system complexity. Such features enhance both the safety and logistical feasibility of propulsion systems utilizing N_2O, making it

an attractive oxidizer for missions where operational simplicity and robustness are prioritized[16,17].

INFLUENCE OF ALUMINUM ADDITIVES ON REGRESSION RATE

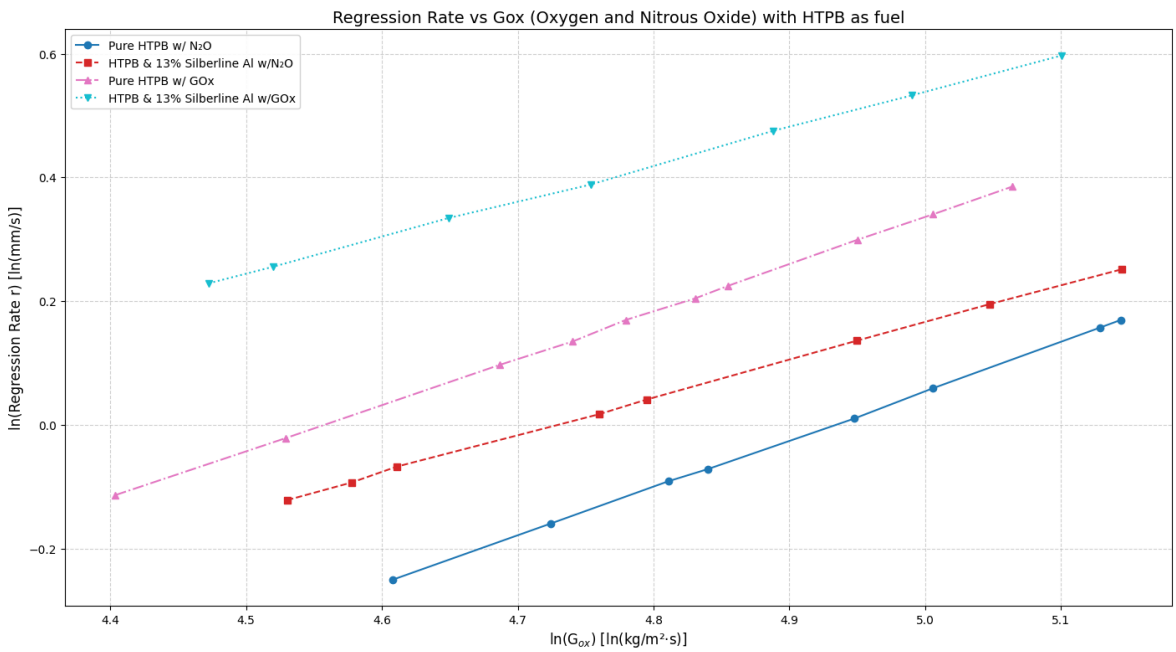

Figure 5.4: Regression rate comparison of HTPB fuel and aluminum (Silberline) additive, using oxidizer as gaseous O_2 and N_2O.[16]

To offset the lower regression rates associated with N_2O, aluminum particles are often added to the HTPB matrix. These particles enhance both convective and radiative heat feedback mechanisms at the fuel surface. Evans et al.[17] reported that the incorporation of 13 wt% nano-sized aluminum flakes increased regression rates by approximately 15% with N_2O and 20% with GOX. This performance gain helps bridge the regression gap between the two oxidizers.

Supporting this, Carmicino et al.[16] found that aluminized HTPB containing 3–12.8 wt% Al exhibited up to 25% higher regression rates at a total mass flux of 100 kg/m^2·s when combusted with N_2O, compared to pure HTPB. Interestingly, the enhancement was more significant in N_2O-fueled systems, potentially due to a greater relative increase in adiabatic flame temperature. However, this benefit diminishes at higher mass fluxes (>150 kg/m^2·s), suggesting a transition to convective heat transfer-dominated regimes where radiative feedback becomes less influential[16].

PARTICLE CHARACTERISTICS AND MECHANICAL INTEGRITY

The effect of aluminum particle size and loading level on regression behavior has been found to be secondary to total energy content. For example, Carmicino et al.[16] demon-

strated that formulations with 3 wt% nano-aluminum (Alex®) and 12.8 wt% micron-sized aluminum (μAl) produced comparable increases in regression rate, indicating that energy density is a more critical parameter than particle morphology or quantity.

Importantly, the addition of aluminum did not compromise the structural integrity or mechanical stiffness of the HTPB grains, preserving their suitability for use in robust motor designs. In contrast, paraffin-based fuels—despite achieving regression rates approximately 80% higher than those of aluminized HTPB—exhibited poor mechanical behavior and lower combustion efficiencies (C^* efficiency ∼60–93%) due to unburned molten fuel droplets and surface sloughing[16].

versatile solution across a wide spectrum of aerospace applications.

Chapter 6

CONCLUSION

Nitrous oxide (N_2O) presents a compelling oxidizer choice for hybrid rocket propulsion, particularly in missions where safety, operational simplicity, and logistical efficiency are paramount. Compared to gaseous oxygen (GOX), N_2O leads to approximately 25% lower regression rates in hydroxyl-terminated polybutadiene (HTPB) due to its inert nitrogen content and associated reduction in flame temperature. However, N_2O offers significant advantages, including self-pressurization at ambient temperatures, non-cryogenic storage, and reduced ground support complexity—making it ideal for sounding rockets, suborbital vehicles, and educational platforms.

Despite its lower performance relative to GOX, the limitations of N_2O can be effectively mitigated through the incorporation of energetic additives. Among these, nano-sized aluminum particles—particularly those with fluoropolymer coatings—stand out for their ability to enhance regression rate and combustion efficiency. These additives improve heat feedback mechanisms via increased radiative and convective transfer, facilitate localized ignition, and contribute to higher flame temperatures, especially at moderate mass flux conditions. At lower to mid-range mass fluxes ($\sim$100 kg/m^2·s), aluminum additives have demonstrated up to 25% improvements in regression rate, narrowing the performance gap between N_2O and GOX systems.

In addition to chemical augmentation, physical parameters such as port geometry, grain length, and surface area-to-volume ratio remain dominant factors in determining regression behavior. The interplay between fuel composition and geometric design necessitates a holistic approach to optimization, particularly in light of deviations from classical boundary-layer models. This underscores the importance of updated empirical correlations and continued theoretical development.

In summary, although N_2O-based hybrid systems inherently trade off some performance for operational benefits, the strategic use of nano-energetic additives and optimized fuel geometry provides a viable path forward. Future advancements—including high-energy metal additives and advanced grain designs—may further elevate the capabilities of N_2O-driven hybrid propulsion, positioning it as a

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